Pretrained Teacher Selection for Reliable Multi-Teacher Distillation in Low-Resource ADMET

Yuzibo
Affiliation to be finalized

Abstract

Low-resource ADMET prediction can benefit from molecular knowledge encoded in physicochemical descriptors and classical teacher models, but many deployment settings still favor lightweight ECFP-only inference. Prior descriptor-teacher distillation improves ECFP-only students in some regression settings, yet fixed distillation weights and jointly learned routing gates are unstable because they do not provide a reliable supervision signal for deciding which teacher to trust. We propose pretrained teacher selection for multi-teacher distillation, a two-stage framework that first trains a standalone teacher selector from cross-fit pseudo-oracle labels and then freezes the selector to route teacher supervision into an ECFP-only AdapterFusion student. The selector uses teacher uncertainty, consensus deviation, validation-derived priors, and descriptor-space coverage features to predict which molecular expert should supervise each sample. Across five regression ADMET endpoints, three train ratios, and three random seeds, pretrained selector routing improves mean test RMSE over base AdapterFusion (5.0889 vs. 5.1841) and fixed descriptor-teacher distillation (5.0889 vs. 5.1621), while approaching the strongest setting-level teacher-selection baseline (5.0792). A calibrated high-resource distillation schedule further narrows this gap to 5.0814 mean RMSE. Mechanism analyses show that teacher selection is predictable: a random-forest reliability probe achieves 0.5617 weighted oracle-teacher accuracy, outperforming majority and setting-prior baselines. These results indicate that selector supervision, not generic adaptive weighting, is a key ingredient for reliable ECFP-only multi-teacher ADMET distillation.

1 Introduction

Low-resource ADMET prediction is difficult because labeled measurements are scarce, noisy, and endpoint-specific. At the same time, molecules have multiple inexpensive sources of predictive information, including ECFP fingerprints, physicochemical descriptors, and predictions from traditional machine learning models trained on these representations [1, 2, 3, 4]. Descriptor-access models can be strong practical predictors, but many deployment settings still favor a lightweight predictor that consumes only fingerprints at inference [5, 6, 7, 8].

Training-time descriptor transfer addresses this mismatch by using descriptor information during learning while keeping the deployed student ECFP-only. A pilot study shows that descriptor reconstruction and descriptor-teacher distillation can improve an ECFP-only AdapterFusion student for low-resource regression ADMET. This follows the broader knowledge-distillation idea of transferring supervision from a stronger or differently informed model into a lighter student [9]. However, fixed distillation strength is not uniformly optimal, and validation-advantage adaptive weighting underperforms fixed strong distillation. This suggests that the unresolved problem is not whether teacher knowledge can help, but when a student should trust each teacher.

This paper studies teacher selection in low-resource ADMET distillation. We ask: how can an ECFP-only student learn which molecular teacher to trust under low-resource supervision, and why do jointly learned routing gates underperform? We propose a two-stage method that first trains a standalone teacher selector from cross-fit pseudo-oracle labels and then freezes that selector to route teacher supervision into an ECFP-only AdapterFusion student.

The contributions are:

- We formulate ECFP-only low-resource ADMET prediction as a selective teacher-selection distillation problem under a training-time teacher, inference-time ECFP-only constraint.
- We show that oracle best-teacher labels are predictable from teacher uncertainty, consensus deviation, validation priors, and descriptor-space coverage, while jointly learned routing gates remain unstable.
- We propose cross-fit selector pretraining plus frozen top-1 routing, and study train-ratio-aware distillation strength as a secondary calibration.
- We evaluate the method on five regression ADMET datasets and analyze why selector correctness, teacher strength, and student usefulness are related but distinct factors.

2 Related Work

Molecular property prediction and ADMET benchmarks. Molecular property prediction has progressed from fingerprint- and descriptor-based models to graph neural networks and pretrained molecular encoders. ECFP fingerprints remain a strong and lightweight baseline [1], while descriptor and tree-based models are widely used in practical cheminformatics [3, 4, 10]. Graph-based models such as D-MPNN, graph transformers, and geometry-aware pretraining improve molecular representation learning in many settings [11, 12, 13, 14]. ADMET benchmarking resources and platforms, including MoleculeNet, TDC, ADMETlab, and ADMET-AI, make these comparisons more reproducible [5, 6, 7, 8]. Our work does not propose a new molecular encoder; it studies how an ECFP-only student can selectively absorb descriptor-access teacher knowledge during training.

Multi-view and auxiliary molecular learning. Recent molecular learning work shows that multiple molecular views and auxiliary tasks can help, but their utility varies by endpoint and data regime. MolFuse studies cross-view fusion for molecular property prediction [15], while MTGL-ADMET and task-specific auxiliary learning show that auxiliary supervision should be selected rather than used uniformly [16, 17]. This motivates our central framing: descriptor, fingerprint, and descriptor-access teachers should not be averaged blindly under low-resource ADMET.

Knowledge distillation and teacher selection. Knowledge distillation transfers information from a teacher into a student [9]. Multi-teacher distillation extends this idea but introduces a selection problem: different teachers can be reliable on different samples. Prior work has explored mixture-of-experts routing and multi-teacher selection [18, 19, 20, 21]. Molecular distillation has also begun to explore cross-view transfer, including MolKD for reaction-to-molecule knowledge transfer [22]. Our contribution is narrower and ADMET-specific: we pretrain a teacher selector from cross-fit pseudo-oracle labels, freeze it, and use it to route multi-teacher supervision into an ECFP-only student.

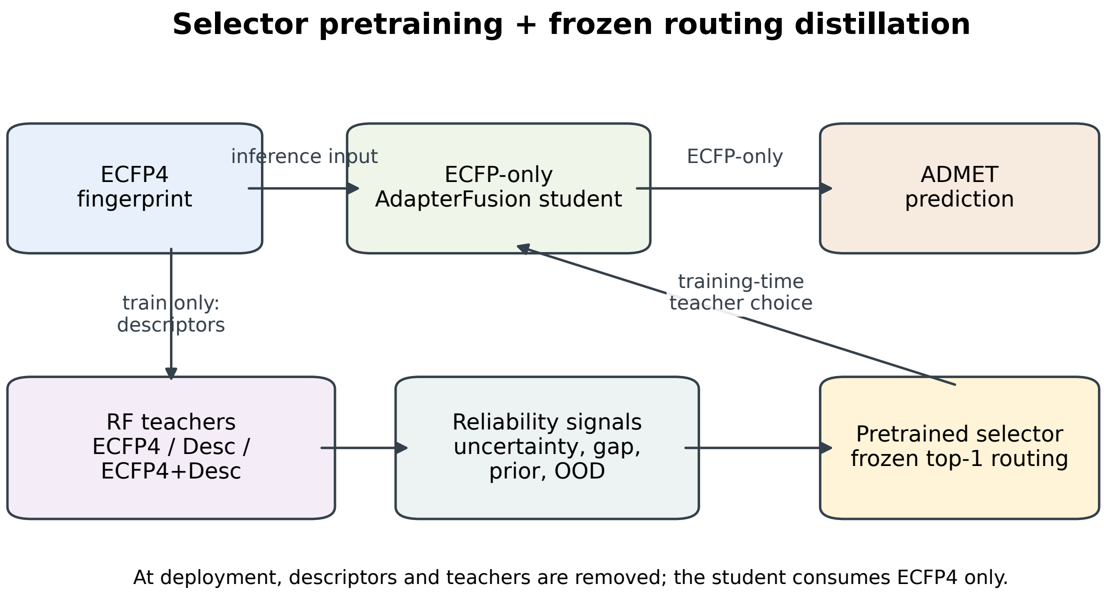

Figure 1: Overview of pretrained teacher selection with frozen routing distillation. During training, RF teachers and reliability signals supervise a standalone selector; the frozen selector routes teacher supervision into an ECFP-only AdapterFusion student. At inference, only ECFP4 is used.

3 Problem Formulation

Let x denote a molecule, $f(x)$ its ECFP4 fingerprint, $d(x)$ its standardized descriptor vector, and y the ADMET target. The student model S_θ is constrained to use only $f(x)$ at inference:

$$\hat{y}_S = S_\theta(f(x)).$$

During training, descriptor information and teacher predictions may be used. Let $T_1, \ldots, T_K$ be molecular expert teachers, where each teacher produces $\hat{y}_i = T_i(x)$. The goal is not to average teacher predictions uniformly. Instead, the student uses a learned teacher selector, motivated by classical mixture-of-experts routing and modern multi-teacher distillation [18, 19, 20, 21],

$$\pi(x) = h_\phi(r(x)),$$

where $r(x)$ contains teacher-selection features derived from teacher uncertainty, consensus deviation, validation priors, and descriptor-space OOD coverage. The selector is trained first, then frozen, and finally used to route teacher distillation while keeping inference ECFP-only.

4 Method

4.1 ECFP-only AdapterFusion student

The student follows an AdapterFusion backbone. An ECFP4 fingerprint is encoded into z_{fp}. A pseudo-descriptor adapter maps this representation into z_{desc}, and a learned gate fuses the two representations for target prediction:

$$z_{fp} = E(f(x)), \qquad z_{desc} = A(z_{fp}),$$

$$g = \sigma(G([z_{fp}, z_{desc}])), \qquad z_{fused} = g \odot z_{fp} + (1-g) \odot z_{desc}.$$

The target head predicts from z_{fused}, while the descriptor head predicts $d(x)$ during training. At inference, the model consumes only ECFP4.

4.2 Molecular expert teachers

The formal experiments use three random-forest teachers that expose different molecular views: `ECFP4_RF`, trained only on ECFP4 fingerprints; `Desc_RF`, trained only on standardized descriptors; and `ECFP4_Desc_RF`, trained on concatenated ECFP4 and descriptor features. These teachers provide a controlled setting in which the student must choose among fingerprint-only, descriptor-only, and descriptor-access supervision.

4.3 Teacher-selection features

For each molecule, the selector receives teacher-specific and shared reliability features. For each teacher T_i, these include RF uncertainty, teacher consensus deviation

$$\left| \hat{y}_i - \frac{1}{K} \sum_j \hat{y}_j \right|,$$

and a validation-derived teacher prior computed from setting-level teacher validation RMSE. A shared descriptor-space OOD distance is appended to the feature vector, following the general QSAR concern that predictions are less reliable outside a model's applicability domain [23, 24]. Teacher-student disagreement is analyzed as a failure mode, but is not part of the formal pretrained selector used in the main experiments.

4.4 Cross-fit selector pretraining

Within each dataset-ratio-seed training split, teachers are re-fit in cross-validation folds to generate out-of-fold train predictions. The pseudo-oracle selector label is the teacher with minimum absolute error on each held-out train sample:

$$y^{sel}(x) = \arg\min_i |\hat{y}_i(x) - y|.$$

A random-forest classifier is trained on these labels with median imputation, 200 trees, maximum depth 8, and minimum leaf size 5. Logistic regression is used only as a diagnostic baseline for selector predictability. The use of out-of-fold labels follows the same leakage-control logic that underlies standard QSAR validation practice [24].

4.5 Frozen routing distillation

After selector pretraining, the selector is frozen. For each training sample, it outputs a probability distribution over teachers. The main method uses hard top-1 routing:

$$i^*(x) = \arg\max_i h_\phi(r(x))_i, \qquad \mathcal{L}_{distill} = (\hat{y}_S - \hat{y}_{i^*(x)})^2.$$

The selector is not updated during student training. This gives the student a fixed teacher-selection signal rather than asking it to learn target prediction and routing from the same low-resource loss.

4.6 Training objective and ratio-aware calibration

For regression, the student is trained with

$$\mathcal{L} = \mathcal{L}_{task} + \lambda_{desc}\mathcal{L}_{desc} + \lambda_{mt} \sum_i w_i(x)(\hat{y}_S - \hat{y}_i)^2.$$

For the main pretrained selector, $w_i(x)$ is one-hot at the selected teacher. For uniform and validation-weighted baselines, $w_i(x)$ is either uniform or derived from teacher validation RMSE. The default setting uses $\lambda_{desc} = 0.1$ and $\lambda_{mt} = 1.0$.

The calibrated selector variant applies a train-ratio schedule:

$$s(r) = \begin{cases} 1.0, & r \in \{10, 20\}, \\ 0.3, & r = 50, \end{cases} \qquad \lambda_{mt}^{eff} = \lambda_{mt} s(r).$$

This high-resource decay tests whether selected teacher supervision should be weakened once more labeled data are available.

5 Experiments

5.1 Datasets and protocol

The main experiments focus on regression ADMET prediction. We evaluate on five endpoints: `caco2_wang`, `lipophilicity_astrazeneca`, `solubility_aqsoldb`, `vdss_lombardo`, and `ppbr_az`. These endpoints are drawn from the TDC and MoleculeNet style molecular property benchmark ecosystem [5, 6]; for solubility, `solubility_aqsoldb` traces to the AqSolDB resource [25]. We choose regression endpoints for the main study to keep the selector target, distillation loss, and evaluation metric within a single metric family. The selected endpoints cover absorption or permeability (`caco2_wang`), physicochemical properties (lipophilicity and solubility), and distribution or binding (`vdss_lombardo` and `ppbr_az`). For each endpoint, we use 10%, 20%, and 50% training ratios with seeds 42, 123, and 3407, giving 45 dataset-ratio-seed settings. The primary metric is test RMSE. Validation splits are used for teacher-prior estimation, model selection, and setting-level teacher baselines, but the pretrained selector's formal supervision comes from cross-fit pseudo-oracle labels within the training split.

5.2 Implementation details

All neural students use ECFP4 fingerprints with radius 2 and 2048 bits as the only inference-time molecular input. The descriptor reconstruction branch uses nine RDKit descriptors: molecular weight, LogP, TPSA, hydrogen-bond acceptors, hydrogen-bond donors, rotatable bonds, aromatic rings, heavy atoms, and ring count. Students are trained with Adam using a learning rate of 5×10^{-4}, weight decay 10^{-5}, batch size 64, at most 200 epochs, and early stopping with patience 30 on validation RMSE. Unless otherwise stated, the descriptor reconstruction weight is 0.1 and the multi-teacher distillation weight is 1.0.

The RF teachers are trained on the same labeled training split as the student and differ only in molecular view: `ECFP4_RF`, `Desc_RF`, and `ECFP4_Desc_RF`. Each RF regressor uses 500 trees and square-root feature subsampling. The formal selector is an RF classifier trained from cross-fit pseudo-oracle labels. Its input contains per-teacher uncertainty, per-teacher consensus gap, per-teacher validation prior weight, and a shared descriptor-space OOD distance. After selector pretraining, selector parameters are frozen during student distillation.

We enforce leakage control at the label-construction stage. For every dataset-ratio-seed setting, selector labels are generated inside the training split using K-fold out-of-fold teacher predictions. Thus, a training sample's pseudo-oracle teacher is determined only by teachers that did not fit that sample in the corresponding fold. Validation and test examples are not used to construct selector supervision; validation predictions are used only for model selection, teacher-prior estimation, and validation-selected baselines.

5.3 Baselines

We compare against several levels of teacher use. The first baseline is the base ECFP-only AdapterFusion student with descriptor reconstruction but no teacher distillation. The second is fixed single-teacher distillation from `ECFP4_Desc_RF`. Multi-teacher controls include uniform averaging, validation-weighted distillation, and `top1_validation`, which selects the best validation teacher for the entire dataset-ratio-seed setting. We also keep the individual RF teachers, `ECFP4_RF`, `Desc_RF`, and `ECFP4_Desc_RF`, as teacher controls.

5.4 Main method

The main method is pretrained selector hard top-1 routing. We additionally report a calibrated variant with high-resource lambda decay, which keeps distillation strength unchanged at 10% and 20% train ratios but reduces it at 50%. Selector-filtered and top-2 variants are treated as secondary comparisons. Selector-confidence auto reweighting is reported only as a diagnostic negative result.

6 Results

6.1 Main regression results

Method	RMSE	Δ Base	Wins	Δ Fixed	Wins	Δ Top1	Wins	Δ Sel.	Wins
Base AdapterFusion	5.1841	–	–	0.0221	15/45	0.1049	11/45	0.0953	11/45
Fixed ECFP4+Desc RF	5.1621	-0.0221	30/45	–	–	0.0828	19/45	0.0732	17/45
Uniform multi-teacher	5.1403	-0.0438	29/45	-0.0217	23/45	0.0611	24/45	0.0514	21/45
Validation-weighted	5.0969	-0.0872	33/45	-0.0651	24/45	0.0177	19/45	0.0081	19/45
Top-1 validation	5.0792	-0.1049	34/45	-0.0828	26/45	–	–	-0.0096	20/45
Pretrained selector	5.0889	-0.0953	34/45	-0.0732	28/45	0.0096	25/45	–	–
Selector + high-resource decay	5.0814	-0.1028	34/45	-0.0807	32/45	0.0021	27/45	-0.0075	9/45

Table 1: Full 45-run regression comparison. RMSE is lower better. Deltas are method minus reference, so negative values indicate improvement.

Table 1 summarizes the full regression matrix. The base ECFP-only AdapterFusion student obtains a mean test RMSE of 5.1841. Fixed single-teacher distillation from the descriptor-access `ECFP4_Desc_RF` teacher improves this to 5.1621, confirming that descriptor-informed teacher supervision can benefit an ECFP-only student. Uniform multi-teacher distillation improves mean test RMSE to 5.1403, while validation-weighted multi-teacher distillation improves it further to 5.0969. The best setting-level baseline is `top1_validation`, which reaches 5.0792 mean test RMSE.

The proposed pretrained selector with frozen top-1 routing reaches 5.0889 mean test RMSE. It improves over base AdapterFusion in 34 of 45 runs and over fixed single-teacher distillation in 28 of 45 runs. Compared with `top1_validation`, the selector wins 25 of 45 runs but remains slightly

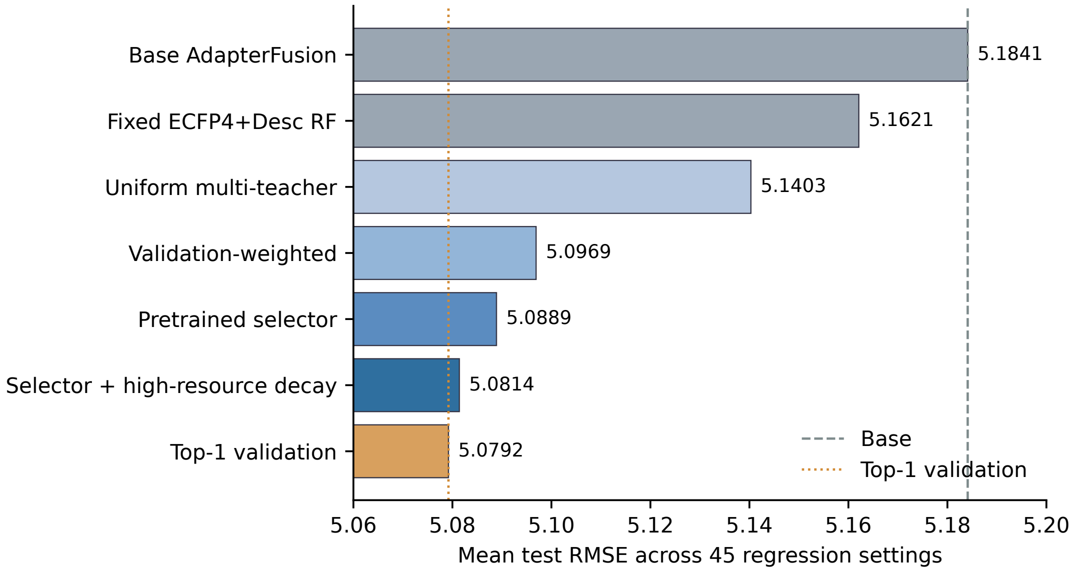

Figure 2: Aggregate mean test RMSE across 45 regression settings. The pretrained selector improves over base AdapterFusion and fixed single-teacher distillation, while high-resource decay nearly matches the strong `top1_validation` baseline.

worse on average by +0.0096 RMSE. Thus, the selector closes most of the gap to the strongest setting-level teacher selector, but it does not decisively dominate that baseline.

Adding high-resource lambda decay further improves the pretrained selector. It reaches 5.0814 mean test RMSE, improves over fixed distillation in 32 of 45 runs, and improves over the plain pretrained selector on mean RMSE by -0.0075. It also wins against `top1_validation` in 27 of 45 runs, although its aggregate mean remains marginally higher than `top1_validation` by +0.0021 RMSE. We therefore treat high-resource decay as a useful calibration of selector routing rather than as a separate main contribution.

6.2 Dataset and train-ratio behavior

The dataset-ratio results show that the benefit of selector routing is not uniform. On `caco2_wang`, the pretrained selector is strongest at 10% and 20% train ratios, while high-resource decay improves the 50% setting relative to the plain selector. The same pattern appears on `ppbr_az`, where the calibrated selector improves the 50% setting from 14.4500 to 14.3472 mean RMSE and beats `top1_validation` at that ratio.

For `lipophilicity_astrazeneca`, pretrained selector routing is strongest at 20%, while the 50% setting is still best served by fixed descriptor-access teacher distillation. For `solubility_aqsoldb`, the differences among uniform multi-teacher, top-1 validation, and selector routing are small. For `vdss_lombardo`, base AdapterFusion remains competitive in the 20% and 50% settings, and high-resource decay slightly weakens the plain selector at 50%. These endpoint differences show that teacher selection is not reducible to a single globally best teacher or a single globally best weighting rule.

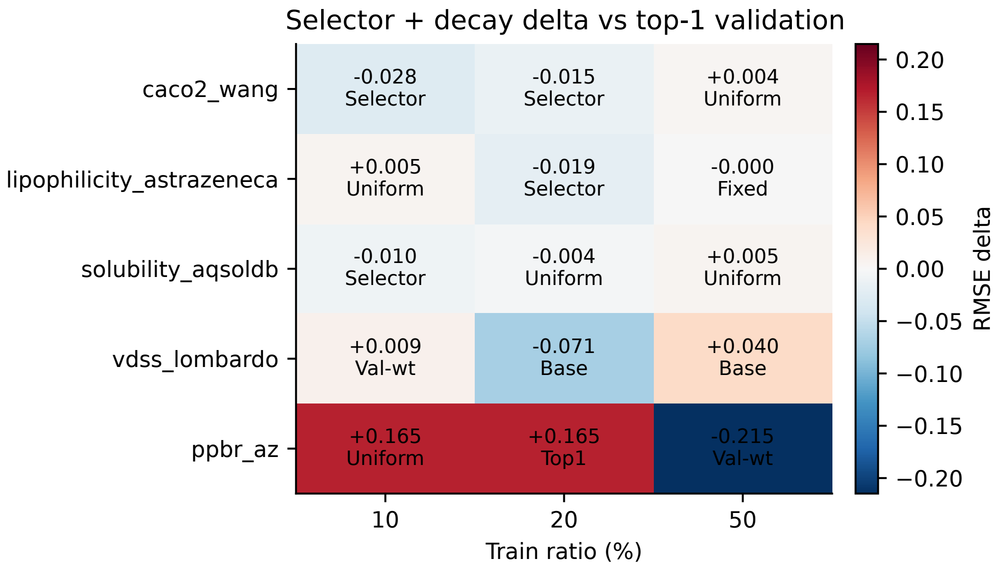

Figure 3: Dataset-ratio behavior of the calibrated selector. Each cell reports the RMSE delta of selector plus high-resource decay relative to `top1_validation`; negative values favor the selector. Text inside each cell shows the best method for that dataset-ratio setting.

Method	Scope	RMSE	Wins	Δ Plain
Plain selector	18 runs	8.6091	–	–
Validation-selected reweight	18 runs	8.6244	5/18	0.0153
Auto confidence reweight	18 runs	8.7875	7/18	0.1783
Low-resource decay	18 runs	8.6666	7/18	0.0575
High-resource decay	18 runs	8.5852	5/18	-0.0239

Table 2: Secondary ablations on the `caco2_wang` and `ppbr_az` partial regression subset. Wins are counted against the plain pretrained selector. Negative deltas indicate lower mean RMSE than the plain selector.

6.3 Negative and secondary ablations

Several alternatives were tested before settling on pretrained selector routing. Table 2 summarizes the main secondary ablations. Joint learned gates and supervised joint gates were weak in smoke experiments, indicating that simply adding a routing network to the student does not provide a stable teacher-selection signal. Selector-confidence reweighting was reproducible and locally useful in specific failure cases, but it was not robust enough to become a main method. A validation-selected policy that chose among no reweighting, global confidence reweighting, and disagreement-only reweighting by validation RMSE improved mean validation RMSE relative to the plain selector (9.6038 vs. 9.6779), but did not improve mean test RMSE (8.6244 vs. 8.6091). The earlier auto-reweight rule was weaker still, increasing mean test RMSE to 8.7875 on the same subset.

The ratio-aware lambda analysis gives a more useful secondary result. Early lambda decay, which reduces distillation strength already at the 20% train ratio, is harmful on the same partial subset. High-resource decay is more defensible because it leaves 10% and 20% settings unchanged and only weakens teacher forcing at 50%. On the full matrix, this improves the plain selector from 5.0889 to 5.0814 mean RMSE.

Diagnostic	Scope	Value
RF selector validation accuracy	45 settings	0.5178
RF selector test accuracy	45 settings	0.5425
RF selector validation macro-F1	45 settings	0.4269
RF selector test macro-F1	45 settings	0.4396
RF gate probe weighted accuracy	45 leave-one-setting-out groups	0.5617
Logistic gate probe weighted accuracy	45 leave-one-setting-out groups	0.5044
Majority-train weighted accuracy	45 leave-one-setting-out groups	0.4818
Setting-prior top-1 weighted accuracy	45 leave-one-setting-out groups	0.4360

Table 3: Selector predictability diagnostics.

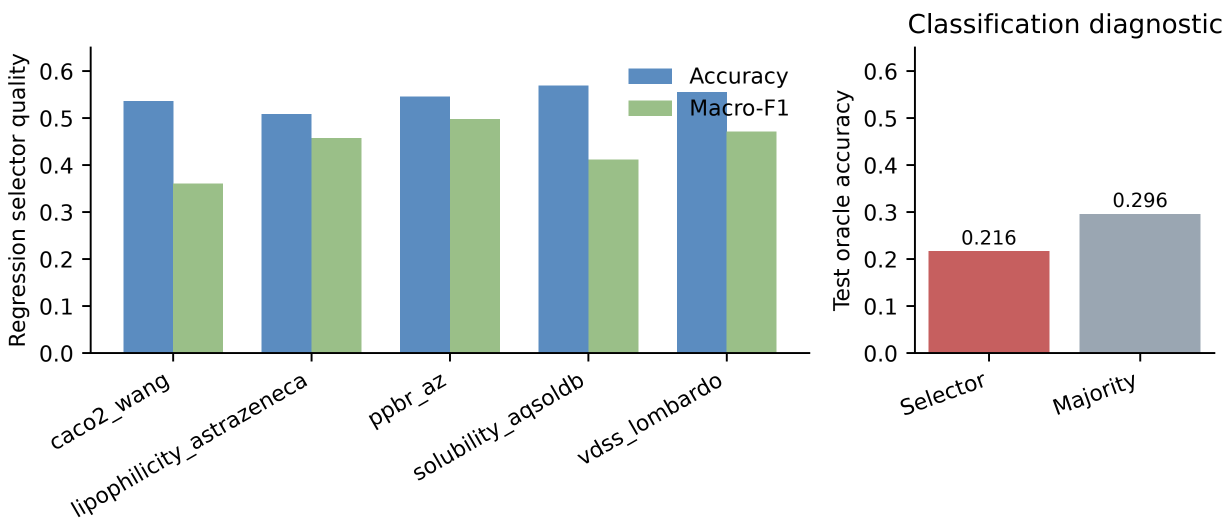

Figure 4: Selector quality diagnostics. Left: regression selector accuracy and macro-F1 by endpoint. Right: the same selector protocol fails on classification diagnostics, where test oracle-teacher accuracy is below the train-majority baseline.

7 Mechanism Analysis

7.1 Teacher selection is predictable

The first mechanism question is whether teacher selection contains learnable signal at all. Table 3 shows that a random-forest gate probe reaches a weighted oracle-teacher accuracy of 0.5617 across 45 held-out settings. This substantially exceeds both the global majority baseline (0.4818) and the setting-level prior baseline (0.4360). A logistic probe also beats these baselines with 0.5044 weighted accuracy, but is clearly weaker than the random-forest probe. The failure of joint gates is therefore not caused by teacher reliability being random or unobservable; rather, the issue is how the teacher-selection signal is supervised and coupled to student optimization.

7.2 Selector quality and failure modes

The pretrained random-forest selector remains noisy, but its quality is stable across the full regression matrix. Averaged over 45 settings, it obtains 0.5178 validation accuracy and 0.5425 test accuracy, with validation and test macro-F1 of 0.4269 and 0.4396. These numbers are not high enough to treat the selector as an oracle, but they are sufficient to route useful supervision into the student.

The remaining gap to `top1_validation` is small but informative. Some settings are still better handled by a coarse setting-level teacher choice. For example, `ppbr_az` at 20% train ratio strongly favors `top1_validation`, while the selector remains weaker despite strong average selector accuracy on that endpoint. This suggests that selector correctness against sample-level pseudo-oracle labels is not always aligned with the student-level RMSE gained from routed distillation.

The high-resource lambda results show that teacher strength matters even after teacher choice is fixed. At 50% train ratio, reducing distillation strength improves the calibrated selector mean RMSE from 4.6001 to 4.5774 across datasets. At 10% and 20%, the calibrated method intentionally matches the plain selector, because early weakening of teacher supervision hurt the partial-regression results.

7.3 Failure modes separate selection from utilization

Setting	Subset	n	Sel. oracle	Top1 oracle	Sel. student wins	Δ abs. err.
`ppbr_az` 20% s42	all	559	0.5546	0.2576	0.3936	+1.0748
`ppbr_az` 20% s42	sel.-better teacher	251	0.9801	0.0000	0.3307	+1.5003
`caco2_wang` 10% s42	all	182	0.5714	0.5989	0.3736	+0.0072
`caco2_wang` 10% s42	sel.-better teacher	17	0.6471	0.0000	0.5294	-0.0126

Table 4: Failure-mode probes comparing teacher-selection correctness with student-level benefit. “Sel.-better teacher” denotes disagreement samples where the selector chooses a lower-error teacher than top-1 validation. Δ abs. err. is selector-routed student absolute error minus top-1 validation student absolute error, so negative values favor the selector-routed student.

Table 4 separates two sources of the remaining gap. On `ppbr_az` at 20% train ratio, the selector is substantially better than top-1 validation at the teacher-selection problem itself (0.5546 vs. 0.2576 oracle accuracy), and on disagreement samples where the selector chooses the lower-error teacher it is nearly perfect. Yet the final student is still worse, with a +1.5003 absolute-error delta on that subset. This is a routing-utilization failure: better teacher choices are not reliably converted into better student predictions.

A direct student-side sanity check supports this diagnosis but also explains why reweighting is not promoted to the main method. On the same `ppbr_az` setting, global confidence reweighting reduces the selector gap to top-1 validation from +0.7527 to +0.3227 test RMSE and turns the sample-level mean absolute-error delta negative. On `caco2_wang`, disagreement-only reweighting improves the smoke test RMSE from 1.7337 to 1.6957. However, selecting a reweight mode by validation RMSE across the 18-run `caco2_wang` + `ppbr_az` subset does not improve test RMSE over the plain selector. Thus, student-side utilization is a real bottleneck, but low-resource validation is not yet sufficient for a stable reweight selection policy.

The `caco2_wang` probe shows the opposite pattern. There, the selector does not beat top-1 validation on teacher-selection accuracy over the full split, so the main bottleneck is selector quality. However, when the selector does choose a better teacher on the disagreement subset, the student effect is directionally positive. These probes support the central interpretation that supervised teacher selection is useful, but future gains require improving both selector quality and the way an ECFP-only student absorbs routed teacher supervision.

8 Discussion

The main lesson from these experiments is that multi-teacher distillation for low-resource ADMET should be treated as a teacher-selection problem, not only as a teacher-averaging problem. Fixed descriptor-teacher distillation improves the ECFP-only student, but it assumes that the same teacher and the same distillation strength are appropriate across endpoints, train ratios, and samples. The multi-teacher baselines show that this assumption is too coarse. This is consistent with recent molecular learning work showing that multiple views or auxiliary tasks can help, but that their value is not uniform across settings [15, 16, 17].

The proposed selector-pretraining strategy addresses a different failure mode from standard adaptive weighting. Joint gates learn routing and target prediction simultaneously, which makes them vulnerable to low-resource task noise and early optimization dynamics. In contrast, the cross-fit selector is trained with an explicit pseudo-oracle teacher-selection target before student training begins. Freezing the selector then turns teacher choice into a stable source of training-time supervision.

At the same time, selector accuracy alone is not the same as student usefulness. A selector can correctly identify the teacher with the lowest sample-level pseudo-oracle error, but the ECFP-only student may not benefit equally from imitating that teacher. The method should therefore be read as evidence that selector supervision is a useful missing ingredient, not as evidence that sample-level routing has fully solved teacher reliability.

Descriptor-access RF teachers remain important controls. In several settings, descriptor-access models or setting-level teacher choices still match or exceed the ECFP-only student. The contribution of this work is not to claim that ECFP-only inference universally dominates descriptor-access prediction. Instead, it targets a deployment constraint: the student can use descriptors and teacher predictions during training, but needs only ECFP4 at inference. This complements graph and pretrained molecular representation learning [11, 12, 13, 14, 26] and recent cross-view molecular distillation work [22] rather than replacing it; those models are natural candidates for future teacher pools once the selector-supervision protocol is established.

9 Limitations

This study has several limitations. First, the main claim is restricted to regression ADMET endpoints. Classification endpoints were explored in pilot work but were less stable, and they should remain supplementary until the teacher-selection protocol is evaluated under classification-specific metrics and calibration behavior. In a five-endpoint classification diagnostic, the same RF selector obtained only 0.2162 mean test oracle-teacher accuracy, below the corresponding train-majority baseline of 0.2958, indicating that the regression pseudo-oracle protocol should not be transferred directly to classification tasks.

Second, the main dataset scope is deliberately restricted to regression endpoints. This avoids mixing RMSE-based selector labels with classification-specific calibration and ROC-AUC behavior, but it means the current claim should be read as regression ADMET evidence rather than a complete ADMET benchmark. Prepared classification endpoints such as blood-brain barrier permeability, human intestinal absorption, P-gp interaction, bioavailability, and hERG inhibition are natural supplementary diagnostics.

Third, the teacher set is deliberately limited. The main experiments use RF teachers over ECFP4, descriptors, and their concatenation. This isolates the teacher-selection problem, but it does not cover the full range of molecular experts available in drug discovery. Adding XGBoost, graph neural networks, or pretrained molecular encoders may change both the attainable student performance and the difficulty of selecting among teachers.

Fourth, the selector features are partly hand-designed. Teacher uncertainty, teacher consensus deviation, validation priors, and descriptor-space coverage are interpretable and effective enough to predict pseudo-oracle teacher labels, but they may not be optimal. Learned reliability representations or meta-learned selector features could improve selection quality.

Fifth, the pseudo-oracle labels are based on teacher absolute error on cross-fit predictions. This avoids direct leakage from fitting and evaluating on the same samples, but it is still an imperfect proxy for the teacher that will most improve the downstream student.

Finally, the calibrated high-resource lambda schedule is heuristic. It is useful because it captures a clear pattern in the experiments, but it is not a fully learned solution to teacher-strength calibration.

10 Conclusion

We presented pretrained teacher selection for reliable multi-teacher distillation in low-resource ADMET. The method separates teacher-selection supervision from student optimization by training a cross-fit pseudo-oracle selector and freezing it for top-1 routing into an ECFP-only AdapterFusion student. Across five regression endpoints, the selector improves over base and fixed distillation and nearly matches a strong setting-level teacher selector. The remaining gap highlights an important distinction: selecting a teacher, choosing how strongly to imitate it, and obtaining student-level benefit are related but distinct problems. This makes supervised teacher selection a practical foundation for future ECFP-only molecular distillation systems.

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